Skills Summary

Similar Triangles: Sides, Perimeter, and Area

Use this reference while completing your worksheet or when studying.

One factor runs the whole picture. If two triangles are similar with scale factor k (new $\div$ old), then every side $\times k$, the perimeter $\times k$, and the area $\times k^2$.

Going backwards: from an area ratio, $k = \sqrt{\text{area ratio}}$.

1. Scale factor and a missing side

Write the similarity statement first — $\triangle ABC \sim \triangle DEF$ pairs A with D, B with E, C with F, so AB pairs with DE.

Then set two matching ratios equal and cross-multiply: $\frac{AB}{DE} = \frac{BC}{EF}$.

Example: $\triangle ABC \sim \triangle DEF$ with $AB = 4$, $BC = 6$, $DE = 10$. Find EF.
Step 1. Two corresponding pairs are known (AB with DE) and one is half missing (BC with EF), so a single proportion settles it.
Step 2. $\dfrac{4}{10} = \dfrac{6}{EF}$, so $4 \cdot EF = 60$.
$EF = \mathbf{15}$

Try it: $\triangle PQR \sim \triangle STU$ with $PQ = 9$, $QR = 12$, $ST = 6$. Find TU.

check: $TU = \mathbf{8}$

2. Perimeter scales by k — not by k^2

A perimeter is a sum of sides, so multiplying every side by k multiplies the total by k as well.

This runs both ways: if you are given the two perimeters, their ratio IS k, even when no single side is known.

Example: Corresponding sides are 5 cm and 8 cm; the smaller triangle's perimeter is 20 cm. Find the larger perimeter.
Step 1. Perimeter is built from sides only, so the side ratio is the factor — no squaring belongs here.
Step 2. $k = \dfrac{8}{5} = 1.6$, so the larger perimeter is 20×1.6.
The perimeter is **32** cm.

Try it: Corresponding sides 6 cm and 9 cm; the smaller perimeter is 22 cm. Find the larger perimeter.

check: **33** cm

3. Area scales by k squared

Area covers two directions at once, so both of them stretch: the area ratio is k^2. With $k = 3$, a triangle's sides triple but its area becomes 9 times as large.

Given the areas instead, undo the square: $k = \sqrt{\text{area ratio}}$, and only then scale a length or a perimeter.

Example: Corresponding sides are 4 cm and 10 cm; the smaller triangle's area is 12 cm^2. Find the larger area.

Step 1. The question asks for an area, and areas scale by the square of the side ratio, so square k before multiplying.

Step 2. $k = \dfrac{10}{4} = 2.5$ and $k^2 = 6.25$, so the area is 12×6.25.

The area is **75** cm^2.

Try it: Corresponding sides 3 cm and 6 cm; the smaller area is 18 cm^2. Find the larger area.

check: **72** cm^2

(!) Watch out: do not scale an area by k, and do not scale a perimeter by k^2. When a problem hands you *areas* and asks for a *length*, take the square root first — that is the step most often skipped.